

Cybersecurity

How Sweet is the Terminal?

Lesson 3.2



Learning Objectives

By the end of today's class, you will be able to:

- 1 Identify and explain the structure of a terminal command.
- 2 Explain how options modify the default behavior of a terminal command.
- 3 Use commands like `head`, `tail`, `more`, and `less` to preview files.
- 4 Combine multiple commands in sequence to accomplish relevant IT tasks.



Welcome

Terminal Review

Commands for navigating a file directory:

<code>pwd</code>	Displays the current working directory
<code>ls</code>	Lists the directories and files in the current directory
<code>cd</code>	Navigates into a directory
<code>cd ../</code>	Navigates out of a directory
<code>clear</code>	Clears the terminal history on the page

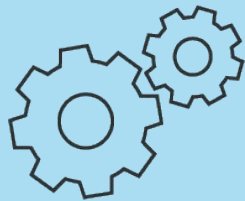
Terminal Review

Commands for making and removing files and directories:

<code>mkdir</code>	Creates a new directory
<code>rmdir</code>	Removes a directory
<code>touch</code>	Creates an empty file
<code>rm</code>	Removes a file

Commands for moving and copying files:

<code>cp</code>	Copies files to a new location
<code>mv</code>	Moves files to a new location



Preview Commands



Security and IT professionals often **preview** the contents of a file.



Security Professional Examples

The **security administrator** needs to preview the top 10 lines of an email.

The **network administrator** needs to preview the bottom of a log file.

```

Current time: 2013-07-23T22:07:56.000 Original time: 2013-07-23T22:07:56.000 Offset: +0.000
Known message fields:
+ log_hostname = Example-iMac.local
+ log_procname = sudo
+ log_pid = 43202
Discovered message fields:
+ col_0 = ghost
+ TTY = ttys006
+ PWD = /Users/ghost/Downloads
+ USER = root
+ COMMAND = /opt/local/bin/port upgrade outdated
Jul 23 22:08:01 Example-iMac com.apple.launchd[1] (com.apple.coremedia.videodecoder[34409]): Exit timeout elapsed (20 se
Jul 23 22:09:33 Example-iMac com.apple.launchd.peruser.502[40247] (com.apple.cfprefsd.xpc.agent[40253]): Exited: Killed:
Jul 23 22:09:33 Example-iMac kernel[0]: memorystatus_thread: idle exiting pid 40253 [cfprefsd]
Jul 23 22:09:34 Example-iMac com.apple.launchd.peruser.502[40247] (com.apple.distnoted.xpc.agent[40252]): Exited: Killed
Jul 23 22:09:34 Example-iMac kernel[0]: memorystatus_thread: idle exiting pid 40252 [distnoted]
Jul 23 22:09:34 Example-iMac com.apple.launchd[1] (com.apple.iCloudHelper[39636]): Exited: Killed: 9
Jul 23 22:09:34 Example-iMac kernel[0]: memorystatus_thread: idle exiting pid 39636 [com.apple.iCloud]
Jul 23 22:09:35 Example-iMac com.apple.launchd[1] (com.apple.hiservices-xpcservice[35049]): Exited: Killed: 9
Jul 23 22:09:35 Example-iMac kernel[0]: memorystatus_thread: idle exiting pid 35049 [com.apple.hiserv]
Jul 23 22:09:35 Example-iMac com.apple.launchd[1] (com.apple.hiservices-xpcservice[34421]): Exited: Killed: 9
Jul 23 22:09:35 Example-iMac kernel[0]: memorystatus_thread: idle exiting pid 34421 [com.apple.hiserv]
Jul 23 22:09:35 Example-iMac com.apple.launchd.peruser.501[237] (com.apple.scopedbookmarksgent.xpc[34399]): Exited: Kil
Jul 23 22:09:35 Example-iMac kernel[0]: memorystatus_thread: idle exiting pid 34399 [ScopedBookmarkAg]
Jul 23 22:09:36 Example-iMac com.apple.launchd.peruser.501[237] (com.apple.accountsd[34365]): Exited: Killed: 9
Jul 23 22:09:36 Example-iMac com.apple.launchd[1] (com.apple.ShareKitHelper[34364]): Exited: Killed: 9
Jul 23 22:09:36 Example-iMac kernel[0]: memorystatus_thread: idle exiting pid 34365 [accountsd]
Jul 23 22:09:36 Example-iMac kernel[0]: memorystatus_thread: idle exiting pid 34364 [com.apple.ShareK]
Jul 23 22:09:36 Example-iMac com.apple.launchd.peruser.89[33810] (com.apple.cfprefsd.xpc.agent[33814]): Exited: Killed:

```

More, Less, Head, and Tail

To preview and scroll through a whole file, use the more or less commands.

more	View a file one page at a time. Space bar moves you to the next page.
less	Similar to more, but it allows more flexibility. You can scroll forward and backward one line at a time, and it supports horizontal scrolling.

To preview a file by a certain number of lines, use the head or tail commands.

head	Displays the top 10 lines of a file
tail	Displays the bottom 10 lines of a file



Modify the Preview

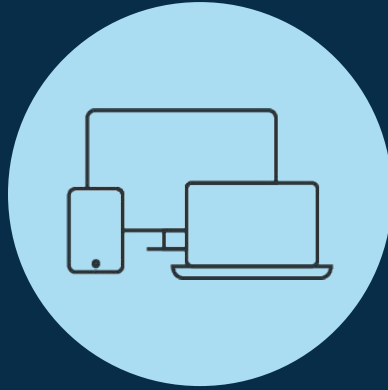
The syntax for head is:

`head` and `tail` commands preview 10 lines by default. This number can be changed easily.

```
head -number file
```

:

```
head -50 logfile.txt
```



Instructor Demonstration

Preview Commands



Activity:

Oh Henry, What Did You Do?

Continue in your role as a security analyst at Candy Corp.

- Your manager identified two employees who may be working with Sugar Corp—Henry and Ruth—and quickly pulled some files from their computers.
- You must preview the files to determine which have readable text data that can be used for analysis.

Suggested Time:
15 Minutes

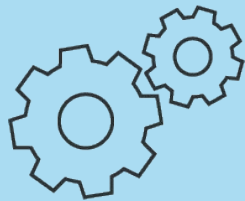


Time's up!
Let's review



Questions?





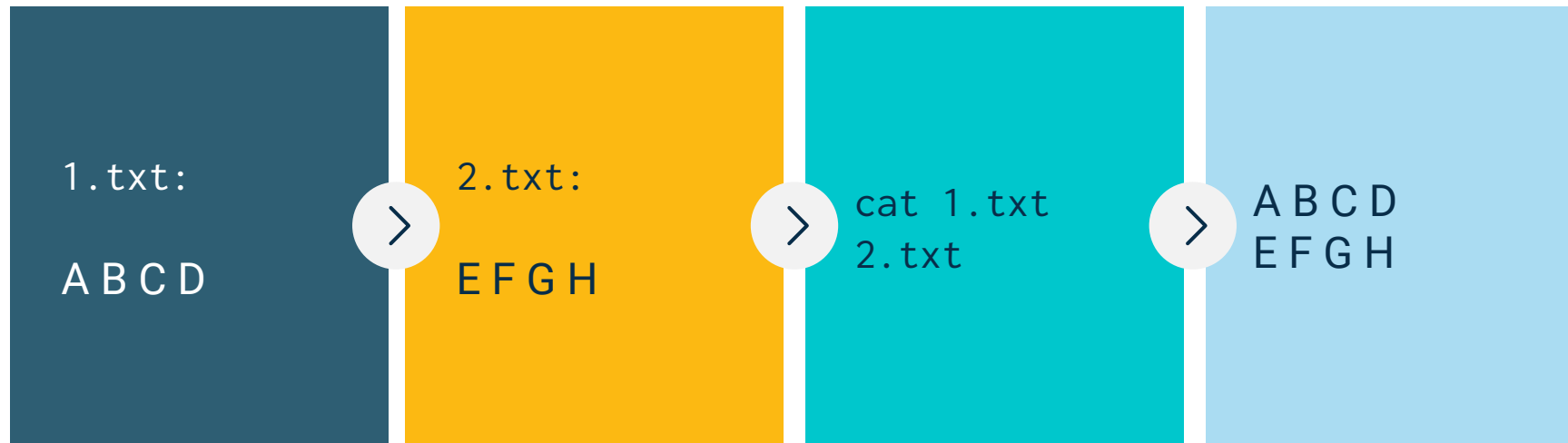
Data Streams and the cat Command



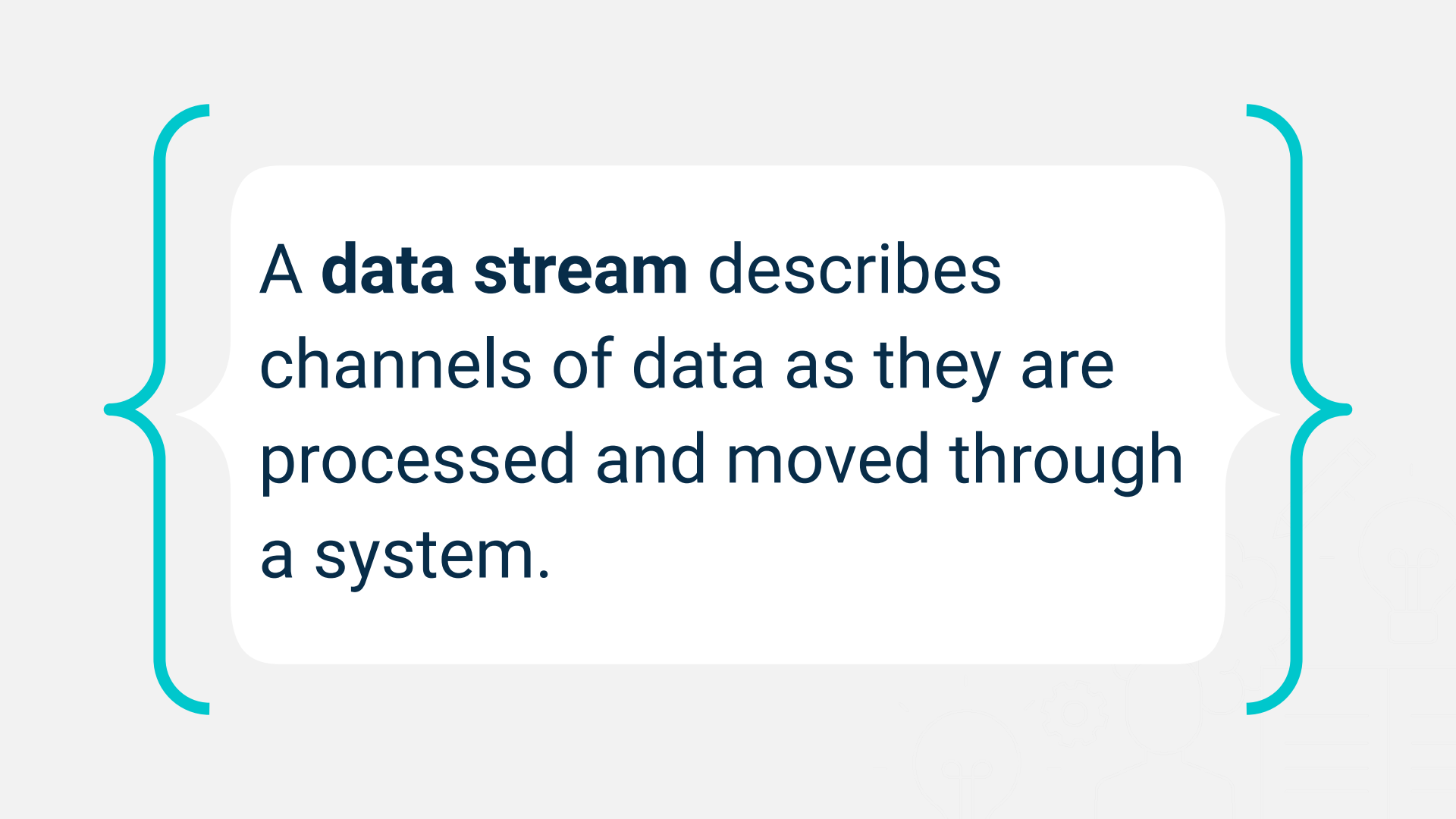
cat (concatenate)
combines files and
displays the results.

The cat Command

Security and IT Professionals often have to combine files



Note: This command **does not** save the results. To save results of the `cat` command, you have to redirect the **data stream**.



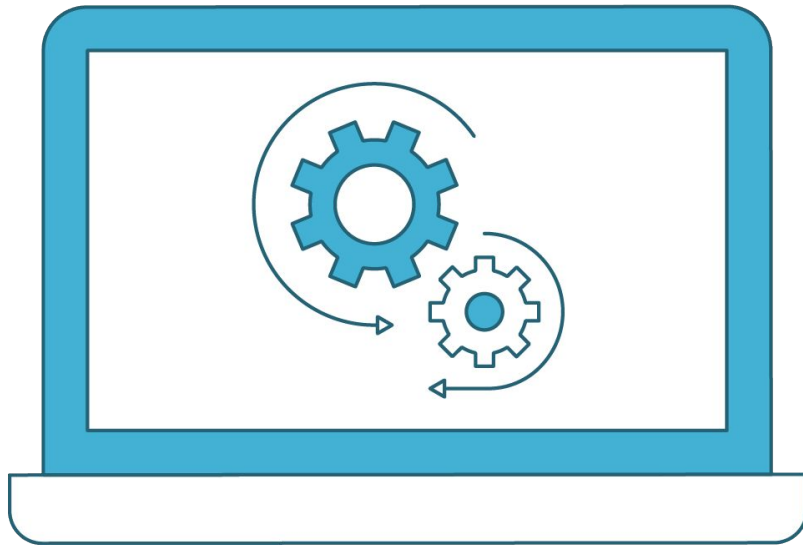
A **data stream** describes channels of data as they are processed and moved through a system.



Data Streams

Move data from one process or program to another

- Most common data streams:
 - `stdin`
 - `stdout`
- Input data streamed using **`stdin`**
- Output data streamed using **`stdout`**



cat with >

```
cat 1.txt 2.txt > combined_1_and_2.txt
```

The command combines two separate files, 1.txt and 2.txt, into a single file called combined_1_and_2.txt.

If combined_1_and_2.txt already exists, it will overwrite the file with the combined data from 1.txt and 2.txt.

cat with >

```
cat 3.txt 4.txt >> combined_3_and_4.txt
```

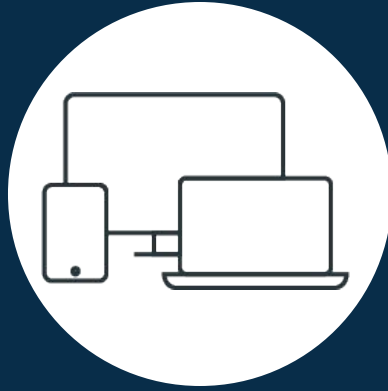
The command combines two separate files, 3.txt and 4.txt, into a single file called combined_3_and_4.txt.

If it already exists, it will append the combined data from 3.txt and 4.txt to the end of the file.

cat and Data Streams

```
cat 1.txt 2.txt > combined_1_and_2.txt
```

- cat takes data directly from stdin and directs it to stdout.
- 1.txt and 2.txt are sent to stdin.
- cat is taking stdin, concatenating the data, and sending it to stdout.
- The result of combining 1.txt and 2.txt is stdout.
- stdout is redirected and written to a file called combined_1_and_2.txt.



Instructor **Demonstration**

cat



Activity:

Internal Investigation: Find the Kit cat Burglar

Continue in your role as a security analyst at Candy Corp.

- Your manager believes Henry and Ruth are sharing secret recipes and collected some files from their computers to help you build your case.
- You must analyze this additional data and prepare a combined evidence file for the local authorities.

Suggested Time:
20 Minutes



Time's up!
Let's review

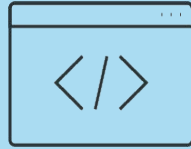


Questions?





Break
15 mins



Command-Line Structure

Command-Line Structure

We're familiar with the following structure:

```
<command> <argument>
```

Example:

```
touch myfile
```

Command

Argument



IT professionals often need to run commands with more **specific parameters**.

Command-Line Parameters

Commands have options that can modify their behavior.

01

Add an option to modify the command's default behavior



02

Add an option and an argument to modify the default behavior on files or directories



03

Add options that require their own arguments, called parameters

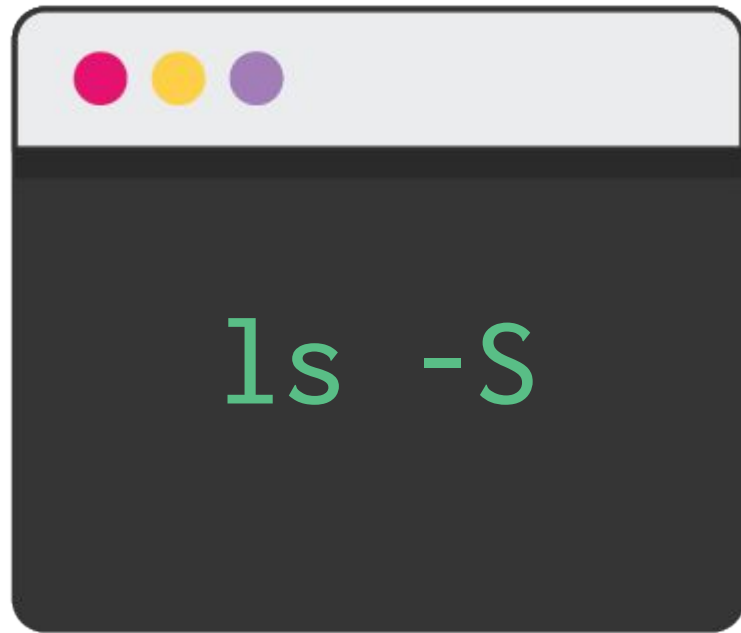


Method 1

Add an option to modify the command's default behaviour.

Use the command `ls -S` to list files by size

- By default, `ls` simply lists the files in the current directory. The `-S` option modifies that behaviour to list by size, largest first.
- The syntax:
 - `ls` is the command.
 - `-S` is the option.

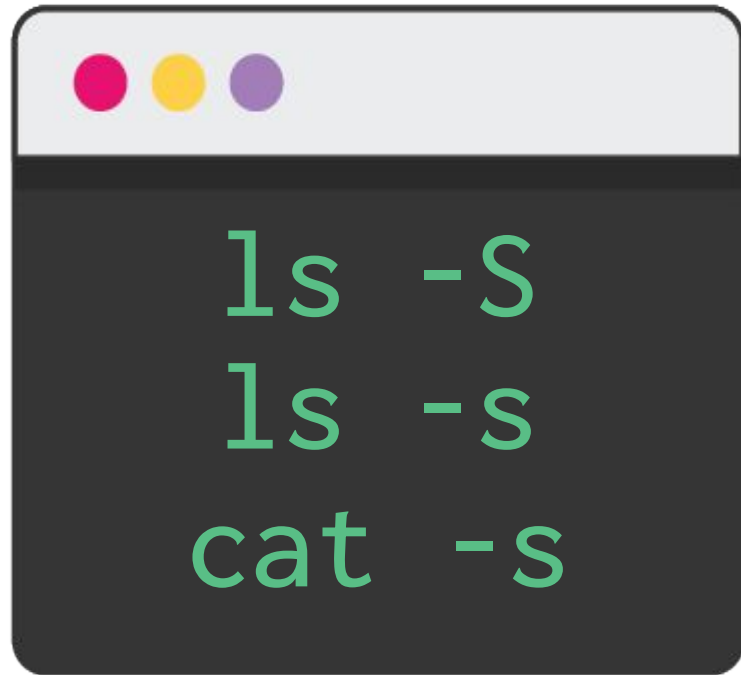


Method 1

Add an option to modify the command's default behaviour.

Keep in mind:

- Options always use a hyphen.
- Options, just like commands, are case sensitive:
 - `-s` and `-S` provide different results.
- Options have different uses for different commands:
 - `ls -s` will print the size of each file.
 - `cat -s` will suppress repeated empty output lines.



Method 2

Add an option and an argument to modify the default behavior.

- This command displays the line numbers of `logfile1.txt`.
- By default, the `cat` command concatenates multiple files or displays the contents of a single file.
- Adding the option `-n` modifies the behavior by displaying the line numbers preceding each line.



Method 3

Add options that require their own arguments, called parameters.

- This command previews the first four lines of `logfile.txt`.
- By default, `head` displays the top ten lines of a file.
- The option `-n` changes the number of lines displayed.
 - `-n` requires a **parameter** specifying the number of lines.
 - Parameters provide additional details on how to modify a command's default behavior.



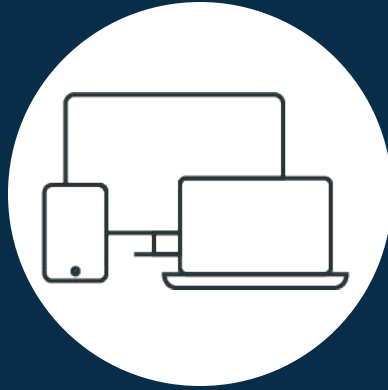


Demonstration Scenario

We are security analysts at ACME Corp who must clean up some evidence files, as server space is low.

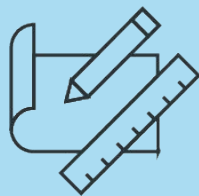
- We need to delete the three largest evidence files, as long as they don't contain the user Sheila.
- The log files are not more than 40 lines each.





Instructor **Demonstration**

Options



Welcome to Man Pages



IT and security professionals can learn and manage options with **man** (manual) **pages**.

Man Pages

Each man page contains:

- Name of the command
- Synopsis (includes syntax)
- Description
- Options and option parameters

The command:

```
man <command>
```

For example:

```
man ls
```

LS(1)

BSD General Commands Manual

LS(1)

NAME

ls -- list directory contents

SYNOPSIS

ls [-ABCFGHLOPRSTUW@abcdefghijklmnopqrstuvwxyz1] [file ...]

DESCRIPTION

For each operand that names a file of a type other than directory, **ls** displays its name as well as any requested, associated information. For each operand that names a file of type directory, **ls** displays the names of files contained within that directory, as well as any requested, associated information.

If no operands are given, the contents of the current directory are displayed. If more than one operand is given, non-directory operands are displayed first; directory and non-directory operands are sorted separately and in lexicographical order.

The following options are available:

- e Display extended attribute keys and sizes in long (-l) output.
- 1 (The numeric digit ``one''.) Force output to be one entry per line. This is the default when output is not to a terminal.
- A List all entries except for . and ... Always set for the super-user.
- a Include directory entries whose names begin with a dot (.).
- B Force printing of non-printable characters (as defined by ctype(3) and current locale settings) in file names as \xxx, where xxx is the numeric value of the character in octal.

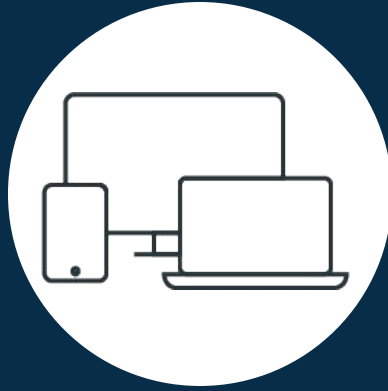


Demonstration Scenario

We are security analysts at ACME Corp and our manager asks you to count the number of server logins on October 13, 2019, as they believe it is higher than usual

- They tell us to use the command `wc` to count the logins on the server login file.
- We have not used the `wc` command before and will need to use the man page to learn how.





Instructor **Demonstration**

Man Pages



Activity:

Learn New Commands

Continue in your role as a security analyst at Candy Corp.

- There has been a network attack on Candy Corp's websites.
- Your manager provides you with log files for each of the websites.
- Each log file contains the IP addresses that were connecting to that website on the day of the attack.
- To determine which website was the main target, you will count the IP addresses in each of these log files.

Suggested Time:
20 Minutes



Time's up!
Let's review



Questions?





Recap

Command-Line Navigation

1 Data streams and the cat command

2 Using `wc` for counting

3 Man pages



Thank you